

Operating temperature of photovoltaic modules: A survey of pertinent correlations

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ABSTRACT

The importance of solar cell/module operating temperature for the electrical performance of silicon-based photovoltaic installations is briefly discussed. Suitable tabulations are given for most of the explicit and implicit correlations found in the literature which link this temperature with standard weather variables and material/system-dependent properties, in an effort to facilitate the modeling/design process in this very promising area of renewable energy applications.

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1. Introduction

The important role of the operating temperature in relation to the electrical efficiency of a photovoltaic (PV) device, be it a simple module, a PV/thermal collector or a building-integrated photovoltaic (BIPV) array, is well established, as can be seen from the attention it has received by the scientific community. A scan of the relevant literature produces dozens of correlations expressing T_c , the PV cell temperature, as a function of the pertinent weather variables, namely, ambient temperature, T_a , and local wind speed, V_w , as well as of the solar radiation flux, G_T . These correlations include, as parameters, material and system-dependent properties such as glazing-cover transmittance, τ , plate absorptance, α , etc. Fig. 1 gives a schematic of the thermal energy exchanges between a rack-mounted PV module and the environment, which involve such variables and parameters. An equally impressive number of correlations can also be retrieved, which express the adverse effect of an operating temperature increase upon the PV module's electrical efficiency (or, equivalently, power).

With regard to the relevant weather variables, and qualitatively speaking, it was found that the PV cell temperature rise over the ambient one is extremely sensitive to wind speed, less so to wind direction, and practically insensitive to the atmospheric temperature [1]. On the other hand, it obviously depends strongly on the

impinging irradiation, i.e. the solar radiation flux on the cell or module.

From the mathematical point of view, the correlations for the PV operating temperature are either explicit in form, thus giving T_c directly, or they are implicit, i.e. they involve variables which themselves depend on T_c . In this last case, an iteration procedure is necessary for the relevant calculation. Most of the correlations usually include a reference state and the corresponding values of the pertinent variables.

2. Implicit correlations for the PV operating temperature

The thermal environment which establishes the instantaneous value of the PV module's operating temperature is quite complex. Aside from internal processes taking place within the semiconductor material during its bombardment by photons – which leads to the production of electricity but also to the release of the non-converted energy as heat – standard heat transfer mechanisms such as convection and radiation must be taken into account in the relevant energy balance on the module, a procedure which leads to the estimation of T_c . In most cases these mechanisms affect both the front and the back side as, in typical installations, provision is usually made to facilitate the removal of the rejected heat so that the module can operate as efficiently as possible. When it comes to free-standing arrays, heat conduction through the mounting frame should also be taken into account, although at steady-state conditions conduction merely transports heat to the surfaces that release it to the ambient by convection and radiation [2].

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Nomenclature		η	cell/module electrical efficiency
G_T	solar radiation flux on module plane (W/m^2)	τ	transmittance of glazing
INOCT	installed nominal operating cell temperature ($^{\circ}\text{C}$)	<i>Subscripts</i>	
k	Ross coefficient – Eq. (4) (Km^2/W)	a	ambient
NOCT	nominal operating cell temperature ($^{\circ}\text{C}$)	b	back side
T_b	back-side cell temperature (K)	c	cell/module
T_c	cell/module operating temperature (K)	h	standoff height
T_a	ambient temperature (K)	L	loss
U_L	thermal loss coefficient ($\text{W/m}^2\text{K}$)	NOCT	at NOCT conditions
V_w	wind speed (m/s)	NTE	nominal terrestrial environment
<i>Greek letters</i>		ref	at reference conditions
α	solar absorptance of PV layer	T	on module's tilted plane
		w	wind induced

The traditional steady-state energy balance which leads to the determination of the operating temperature of the PV cell/module requires as input

- thermal and physical properties of the cell/module;
- solar resource and weather data;
- heat transfer coefficient due to the wind.

The latter is not easy to determine, especially in the field, where monitoring the wind or establishing uniform conditions for the relevant measurements is a formidable job. This explains the plethora of wind driven heat transfer coefficient correlations which have appeared in the literature in recent years (cf. [3]).

The temperature of the cells within a PV module, i.e. T_c , may be higher than the back-side temperature, T_b , by a few degrees, and their difference depending on the module substrate materials and on the solar radiation flux levels. A simple expression relating the two temperatures is

$$T_c = T_b + \frac{G_T}{G_{\text{ref}}} \Delta T \quad (1)$$

in which G_{ref} is a reference solar radiation flux on the module (1000 W/m^2), and ΔT is the temperature difference between the PV cells and the module back side, at this reference solar radiation flux [4].

An established procedure to formulate the PV cell/module operating temperature involves use of the so-called nominal operating cell temperature (NOCT), defined as the temperature of a device at the conditions of the nominal terrestrial environment (NTE): solar radiation flux (irradiance) 800 W/m^2 , ambient temperature 20°C , average wind speed 1 m/s , zero electrical load (i.e. open circuit), and free-standing mounting frame oriented “normal to solar noon” [5,6]. With symbols, $\text{NOCT} = (T_c - T_a)_{\text{NTE}} + 20^{\circ}\text{C}$. The relevant method, which assumes that both sides of the module “feel” the same ambient temperature, is based on the fact that – as noted in Section 1 – the temperature difference ($T_c - T_a$) is practically independent of T_a but linearly proportional to the incident solar radiation flux. In addition, it makes the approximation that the overall heat loss coefficient, U_L , is constant.

This last approximation, however, is not really necessary, as it does not lead to a substantial simplification. Thus, if the energy balance for a unit module area, namely,

$$(\tau\alpha)G_T = \eta_c G_T + U_L(T_c - T_a) \quad (2)$$

is written for NOCT conditions (i.e. with $\eta_c = 0$) and the resulting equation is combined with Eq. (2) by dividing the two, an expression for T_c can be easily obtained in the form

$$T_c = T_a + \left(\frac{G_T}{G_{\text{NOCT}}} \right) \left(\frac{U_{L,\text{NOCT}}}{U_L} \right) (T_{\text{NOCT}} - T_{a,\text{NOCT}}) \left[1 - \left(\frac{\eta_c}{\tau\alpha} \right) \right] \quad (3)$$

Here η_c is the electrical efficiency of the PV module and, for simplicity, we have written G_{NOCT} and T_{NOCT} instead of $G_{T,\text{NOCT}}$ and $T_{c,\text{NOCT}}$, respectively [7, p. 760]. But η_c is itself a function of T_c , as pointed out in Section 1. Therefore, Eq. (3) is an implicit equation for the PV module temperature, suitable mainly for situations in which the modules are mounted in the free-standing manner described above. That is, one should avoid using it for BIPV installations, where the two sides of the modules are subjected to quite different environmental conditions and, thus, require modified prediction approaches (the NOCT model here can underpredict T_c by as much as 20 K [8]).

In BIPV situations, the PV modules are mounted at an optimized distance from the façade of the building and, therefore, the energy balance does not limit itself to the module layer(s). Instead, it includes the air layer in the gap between module and wall, and the wall itself. Thus, Eq. (2) is typically replaced by a system of three simultaneous equations, each one resulting from an individual energy balance on the respective layer and featuring the respective temperature, i.e. that of the PV module, of the gap air, and of the wall. Needless to say, such balances consider all modes of heat transfer among the layers, the ambient, and (at times) the interior space and obviously can vary widely in the detail. Quite often the methodology adopted involves a lumped analysis approach, typically assuming uniform conditions across the gap, i.e. bulk flow, while, in more detailed studies, dynamic models and CFD methods are employed [9–11].

In addition to Eq. (3), a number of other implicit equations for T_c found in the literature are also listed in Table 1. Some of them require extra information, i.e. beyond what is provided by the module manufacturer.

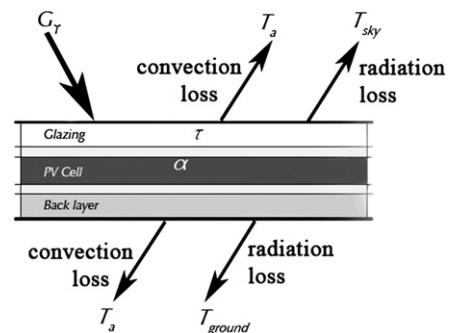


Fig. 1. Simple schematic of the thermal processes in a rack-mounted PV module.

Table 1
Implicit equations for T_c

Correlation	Comments	Ref.
$T_c = T_a + [(\alpha - \eta)G_T + (a + bT_a)] / (17.8 + 2.1V_w)$	a, b are empirical functions of PV and ground emissivities ε_{PV} and ε_g , respectively, and of a cloudiness factor, ε_c^a	[30]
$T_c = T_a + \alpha G_T (1 + \beta T_a) (1 - \gamma v_w) (1 - 1.053 \eta_c)$	$v_w = 1$ m/s, G_T in W/m^2 , $\alpha = 0.0138$, $\beta = 0.031$, $\gamma = 0.042$, T in $^\circ C$	[31]
$T_c = T_a + G_T \left(\frac{\tau \alpha}{U_L} \right) \left[1 - \left(\frac{\eta_c}{\tau \alpha} \right) \right]$	$\tau \alpha / U_L$ taken as constant, used in TRNSYS Type 170 – Mode 1 ^b	[32,33,25]
$T_c = T_a + \frac{(\tau \alpha - \eta_c)}{U_L} G_T$	Adapted from Eq. (3), similar to Refs. [32,33], $\tau \alpha / U_L$ determined experimentally	[34]
$T_c = T_a + \frac{G_T}{C_1 + C_2 V_w} \left[(1 - \rho)(1 - \eta) - \frac{\varepsilon T_c^4 - T_{sky}^4}{G_T} \right]$	ρ = module reflectivity, ε = module emissivity, neglects radiative loss to ground, ^c C_1, C_2 conductive and convective coefficients	[35]
$T_c = T_a + \frac{G_T}{G_{NOCT}} (T_{c,NOCT} - T_a) \left(1 - \frac{\eta_c}{\tau \alpha} \right)$	Eqs. (23.3.2) and (23.3.3) of Ref. [32]; assumes constant U_L	[36,32]
$T_c = T_a + \frac{G_T}{G_{NOCT}} (T_{c,NOCT} - T_{a,NOCT}) \left(1 - \frac{\eta_c}{\tau \alpha} \right)$	NOCT ^d conditions: 800 W/m^2 , AM1.5, 20 $^\circ C$, $V_{wind} > 1$ m/s; assumes constant U_L	[6,8]
$T_c = T_a + G_T \left(\frac{\alpha}{U_L} \right) \left[1 - \left(\frac{\eta_c}{\tau \alpha} \right) \right]$	$U_L = h_{radn} + h_{conv}$	[37]
$T_c = T_a + \left(\frac{G_T}{G_{NOCT}} \right) \left(\frac{U_{L,NOCT}}{U_L} \right) (T_{NOCT} - T_{a,NOCT}) \left[1 - \left(\frac{\eta_c}{\tau \alpha} \right) \right]$	$U_L = h_{radn} + h_{conv} + \text{conduction through mounting frame}$	[7, p. 760]
$T_c = T_a + \left(\frac{9.5}{5.7 + 3.8 V_w} \right) \left(\frac{G_T}{G_{NOCT}} \right) (T_{NOCT} - T_{a,NOCT}) \left[1 - \left(\frac{\eta_c}{\tau \alpha} \right) \right]$	Uses Nusselt–Jurgess correlation for convection; $\tau \alpha \approx 0.9$	[7, p. 760]
$T_{bs} = \frac{h_{p1} (\tau \alpha)_{eff} G_T + U_{tb} T_a + h_b T_w}{U_{tb} + h_b}$	T_{bs} = back surface of tedlar, U_{tb} = glass to tedlar coefficient, h_T = tedlar heat transfer coefficient, h_{p1} = penalty factor (resistance due to EVA, Si material, tedlar), ($\tau \alpha$) _{eff} = function of $\eta_c(T_c)$	[38]
$T_c = \frac{\tau [\alpha_c p + \alpha_T (1 - \beta_c)] G_T - \eta_c G_T \beta_c + U_t T_a + U_T T_{bs}}{U_t + U_T}$	U_t = cell to ambient via glass, U_T = tedlar side, p = packing factor, T_{bs} = function of $\eta_c(T_c)$	[39]
$T'_c = \frac{\tau [\alpha_c p + \alpha_T (1 - \beta_c)] G_T + (U_t + U_T) T_a}{U_t + U_T}$	T'_c = cell T at zero water flow, $\tau \alpha_c$ = function of $\eta_c(T_c)$, p = packing factor, $U_{T_s} = \left[\frac{h_T}{k_T} + \frac{1}{h_b} \right]^{-1}$	[38]

Notes:

- General Electric model for residential arrays: 4th degree equation for T_c [40].
- Refined algorithms for T_c prediction within the NOCT methodology are given in Ref. [41].
- In PV/T collectors T_c depends on the inlet temperature, T_i , and at zero flow $T_c = T_{stagnation}$ [42].
- NIST 1-D trans. model: treats BIPV module as a composite, leads to set of non-linear non-homogeneous DEs [8].
- A transient thermal PV model is given in Ref. [43].

^a $a = 208\varepsilon_{PV} + 297.14\varepsilon_a - 594.3\varepsilon_g$ and $b = 6\varepsilon_{PV} + \varepsilon_a - 2\varepsilon_g$.^b The TRNSYS Type 170 – Mode 2 expression, on the other hand, is $T_c = \frac{h_T T_a + h_b T_{room} + G_{total} \tau \alpha_{eff} - G_{rad-loss}}{h_T + h_b}$.^c The loss-to-ground term is $\sigma \frac{\varepsilon T_c^4 - \varepsilon_g T_g^4}{G_T}$.^d NOCT determined via ASTM E1036M – Annex A.1 [6]. All correlations involving NOCT apply to free-standing arrays only.

3. Explicit correlations for the PV operating temperature

The simplest explicit equation for the operating temperature of a PV cell/module links T_c with the ambient temperature and the incident solar radiation flux [12]:

$$T_c = T_a + k G_T \quad (4)$$

In this linear expression, which holds for no electrical load and no wind, the dimensional parameter k , known as the Ross coefficient, is given by the ratio $\Delta(T_c - T_a)/\Delta G_T$, i.e. it is the slope in the $(T_c - T_a)$ versus G_T plot. Earlier reported values for k were in the range 0.02–0.04 Km^2/W [13], but a more recent IEA study extended this range upwards, categorizing the results qualitatively according to level of integration and size of air-gap (if any) behind the modules [14]. Table 2 lists values of parameter k for various array types/mounting schemes, estimated from the plots shown in the above study.

Another simple explicit equation for the solar cell/module's operating temperature can be derived through simplification of

Eq. (3). If the usual assumption of constant U_L is made, then Eq. (3) can be written as [15, p. 88]

$$T_c = T_a + \left(\frac{G_T}{G_{NOCT}} \right) (T_{NOCT} - T_{a,NOCT}) \quad (5)$$

since the term $(\eta_c/\tau \alpha)$ is small compared to unity (cf. [7, p. 760]).

A listing of other explicit equations for T_c found in the literature is given in Table 3. A few of them are in the form of Eq. (4), while others are much more complicated.

The increasing interest in BIPV applications brought forward the need for a proper estimation of NOCT which would take into account the integration-dependent deviation from NTE conditions, namely, the angle of incidence usually being 90° instead of 0°, and the module temperature being higher due to lack of proper cooling from the poorly ventilated back side. Thus, the module's NOCT, which depends on the mounting scheme for a given irradiation level (cf. Fig. 2), must be measured in a properly designed and well controlled outdoor test bed, like the European Commission's test reference environment (TRE) rig that was recently set up at the JRC Ispra¹ for BIPV testing [16]. An earlier effort to measure the effect of the back-side air-gap size upon the temperature of various types of PV modules led to the construction of a special dual test facility consisting of a control and an experimental test bed. Built in New Mexico for the U.S. Department of Energy (DOE), both test beds had identical tiltable roofs and adjustable standoff height mounting

Table 2
Values of parameter k in Eq. (4)^a

PV array type	k (Km^2/W)
Well cooled	0.02
Free standing	0.0208
Flat on roof	0.026
Not so well cooled	0.0342
Transparent PV	0.0455
Façade integrated	0.0538
On sloped roof	0.0563

^a Adapted from data in Ref. [14].

¹ In the context of an energy rating procedure, a site and mounting specific temperature has been also postulated at Ispra, i.e. the normal operating specific temperature (NOST) [76].

Table 3
Explicit equations for T_c

Correlation	Comments	Ref.
$T_c = T_a + kG_T$	$k = \frac{\Delta T_c}{\Delta G_T}$: 0.02–0.04 °Cm ² /W, ^a no load, no wind, T in °C	[12,13,44–47]
Same as above	0.02 < k < 0.056 for BIPV situations. Actual values depend on level of integration and (ventilation) gap size	[14]
Same as above	$k = 0.03, 0.012, 0.0058$ for conventional, and upper or lower module in packaged home system, respectively. T in °C	[48]
$T_c = 3.12 + 0.025G_T + 0.899T_a - 1.30V_w$	18 kW DC output supplying a UPS	[49]
$T_c = 3.81 + 0.0282G_T + 1.31T_a - 1.65V_w$	104 kW array with MPPT	[49]
$T_c = \left[\frac{(U_0/F_p) - (m_0/F_p)(T_a + T_{co})}{2(m_0/F_p)} \right] \left[(1 + Y)^{1/2} - 1 \right]$	Zero subscripts denote annual averages. F_p , packing factor; m_0 , a meteo parameter. Y is similar to left bracket expression involving G_T , T_a and a modified reference efficiency. ^b T_{co} is an annual average T_c	[50]
$T_c = T_a + 0.028G_T - 1.0$	For $1.00 < V_w < 1.5$ m/s, $0 < T_a < 35$ °C	[30]
$T_c = T_a + 0.035G_T$	Typical free-standing earlier modules, $V_w \approx 1$ m/s	[51]
$T_c = T_a + \alpha G_T (1 + \beta T_a) (1 - \gamma V_w)$	Array T , $\alpha = 0.0138$, $\beta = 0.031$, $\gamma = 0.042$, T in °C,	[31]
$T_c = \frac{T_a + \left[\frac{\alpha_{cell} - \eta_{ref} (1 + \beta T_{ref})}{1 - \eta_{ref} \beta \left(\frac{\alpha}{\pi} \right)} \right] \left(\frac{G_T}{\pi} \right)}{1 - \eta_{ref} \beta \left(\frac{\alpha}{\pi} \right)}$	open-circuit version of Table 1 equation	[52]
$T_c = \frac{[(\tau\alpha) - \eta_{ref}]G_T + T_a [h_{cf} + 2\sigma\epsilon_c\tau_{IR}(1 + \cos\beta)T_{sky}T_a^2 + h_{cb} + 4\sigma T_a^2 F_e F_b]}{h_{cf} + 2\sigma\epsilon_c\tau_{IR}T_a^2(1 + \cos\beta) + h_{cb} + 4\sigma T_a^2 F_e F_b}$	α_{cell} = fraction absorbed, $0.004 < \beta < 0.006$ °C ⁻¹ , $h \sim 8.5 + 3.2V_{wind}$, includes both radiation and convection	[53]
$T_c = T_a + 0.0155G_T + 0.7$	h_{cb} , F_e , and F_b refer to the back side and are estimated via tabulated correlations ^c	[54]
$T_c = 30.006 + 0.0175(G_T - 300) + 1.14(T_a - 25)$	PV cells sealed in gas-filled glass tube-flasks. T in °C	[55]
$T_c = T_a + \frac{G_T}{G_{T,ref}} [0.0712V_w^2 - 2.411V_w + 32.96]$	Module T for pc-Si, T in °C	[56]
$T_c = T_a + \frac{G_T}{G_{NOCT}} (T_{c,NOCT} - T_{a,NOCT}) = T_a + \frac{G_T}{800} (NOCT - 20)$	c-Si in open rack mount, T in °C; $V_w < 18$ m/s	[15 (p. 88), 51, 57–62]
$T_c = T_a + (219 + 832\bar{K}_t) \frac{NOCT - 20}{800}$	NOCT: 800 W/m ² , AM1.5, 20 °C, $v_w > 1$ m/s, assumes constant U_L	[63]
$T_c = 30 + 0.0175(G_T - 150) + 1.14(T_a - 25)$	$\bar{K}_t$ = monthly clearness index at optimum tilt, ^d NOCT = 45 °C for c,pc-Si, 50 °C for a-Si, T in °C	[64]
$T_c = T_i + k\eta_T G_T$ $k = \frac{1 - F_b}{F_b U_L}$	Module T for a-Si, T in °C	[42]
$T_c = T_a + \frac{G_T}{G_{T,ref}} \{ T_1 e^{bV_w} + T_2 + \Delta T_{ref} \}$	PV/T system, η_T = thermal efficiency, T_i = tank inlet temperature	[65]
$T_c = T_a + G_T \left[e^{(a+bV_f)} + \frac{\Delta T_{ref}}{G_{T,ref}} \right]$	T_1 = max T at low wind, T_2 = min T at high wind, $\Delta T_{ref} = T_c - T_{back}$ at $G_{T,ref} = 1000$ W/m ² , b = rate of T drop with wind, T in °C	[4]
$T_c = T_a + 0.031G_T$	Used in the SNL model, $\Delta T_{ref} = T_c - T_{back}$ at $G_{T,ref} = 1000$ W/m ² , T in °C, T_m is the back-side temperature, V_f = free-stream wind speed. Dimensionally inconsistent equation!	[66,67]
$T_c = T_a + 0.031G_T - 0.058$	T_c average back and front, T in °C	[68]
Same as in Ref. [65] but for monthly average $T_{c,eff}$	T_c average back and front, T in °C	[69]
$T_c = \frac{U_{PV}T_a + G_T[(\alpha\tau) - \eta_{ref}(1 + \beta T_{ref})]}{U_{PV} - \beta\eta_{ref}G_T}$	Based on meteo data, T in °C	[70]
$T_c = 0.943T_a + 0.028G_T - 1.528V_w + 4.3$	U_{PV} = 2-sides heat transfer coefficient, $\alpha\tau = 0.9$, $U_{PV} = 28.8$ W/m ² K, $U_{PV} = 24.1 + 2.9V_w$	[71]

Notes:

- In PV/T collectors T_c depends on T_i and, at zero flow, $T_c = T_{stagnation}$ [42].
- Ref. [74] presents a two-dimensional dependence for the solar cell temperature, $T_{sf}(x,y)$.
- A rule of thumb for system design is: (average daily cell temperature) = (average daytime ambient temperature + 25) [75].

^a A much higher value (0.078) is reported in Ref. [72].

^b $Y = \frac{4(m_0/F_p)[(U_0/F_p)T_a + (\alpha - \eta_{ref})G_T]}{[(U_0/F_p) - (m_0/F_p)(T_a + T_{co})]^2}$, $\eta_{ref} = \eta_{ref}[1 + a_1 T_{ref} + a_2 \ln(G_T/1000)]$, $a_1 = 0.005$, $a_2 = 0.052$ for Si cells.

^c For zero heat loss from the back side (e.g. in direct-mount situations) $h_{cb} = F_e = 0$ and the expression gives $T_{c,max}$ [73].

^d That is, latitude – declination. For non-optimal values, use a multiplier $C_f = 1 - 1.17 \times 10^{-4}(S_M - S)^2$ (S_M = optimal tilt angle, S = actual tilt angle, in degrees).

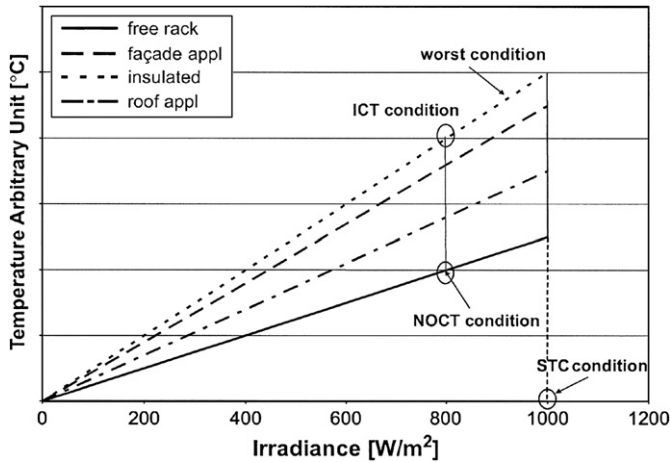


Fig. 2. BIPV mounting induced temperature difference from NOCT as a function of irradiance [16].

capability, i.e. they could vary the gap size. The ventilation height effect on the module's operating temperature was then depicted by a correlation of the form

$$(T_{c,h} - T_a) = a + b(T_{c,h=0} - T_a) \quad (6)$$

in which the subscript h denotes the gap height ($h=0$ refers to a direct-mount array) and a, b regression coefficients, properly normalized to eliminate construction differences between the two beds [17].

4. Operating temperature handling in major PV software tools

It is clear that any simulator of a PV array performance needs the cell/module operating temperature in order to translate the performance of the modules from the standard rating temperature of 25 °C to the modules' performance at operating temperatures. Most of the PV simulation packages base the PV module's description on a solar cell's DC model which has its origins in solid state theory, but which they usually supplement with additional parameters in order to augment its accuracy and make it useful for engineering analysis. The basic equation, which describes the I - V curve for the PV cell, is normally altered in such a way that the computer can derive its own curve-fitting constants from the experimental data input [18]. The scale up is suitably accommodated, with the user inputting the number of parallel strings of modules and the number of modules in a string, and the temperature related input typically includes the relevant coefficients as well as the value of T_c at reference conditions, e.g. NOCT, which are available in the module brochure.

In the PVFORM hourly simulation package, which was based on the rigorous but quite complex model developed earlier at the U.S. Sandia National Laboratory (SNL), the relevant algorithm predicts the cell temperature of a photovoltaic array to within 5 °C, even though it requires a minimum amount of input. The major input parameter to this model is the "installed" nominal operating cell temperature (INOCT), which can be estimated from the traditional NOCT and the mounting configuration, or from cell temperature data of a fielded array² [19]. Table 4 gives the formulas and data for the explicit estimation of INOCT (from the value of NOCT) for

Table 4
Calculation of INOCT from NOCT [19]

PV array mount type	INOCT (°C)
Rack mount	NOCT – 3 °C
Direct mount	NOCT + 18 °C
Standoff/integral	NOCT + X
h^a (cm)	X (°C)
2.5	11
7.5	2
15	–1

^a h is standoff, entrance, or exit height/width, whichever is minimum. If channeled, 4 °C is added.

mounting situations ranging from direct-mount to rack-mount PV systems, including three intermediate standoff heights.³ A simpler empirically based thermal model was later developed at SNL and has proven to be very adaptable and entirely adequate for system engineering and design purposes, producing the expected module operating temperature with the same accuracy [4]. The model, whose equation for T_c is included in Table 3, makes use of Eq. (1) and is utilized in the current successor package, PV Design Pro. In the latter, the user needs to only select the type of module from the program's database, and input the number of parallel connections and series strings of similar panels. Various chart outputs, such as the typical I - V - P , are available to the designer to estimate performance [20]. PVFORM's photovoltaic performance model is also used in PVWATTS, a U.S. National Renewable Energy Laboratory (NREL) produced online calculator/program, which estimates monthly and annual electrical energy production and cost savings for a grid-connected c-Si system via hour-by-hour calculations. The INOCT and the temperature coefficient for power are system parameters, not to be changed by the user of the program [21].

The Photovoltaic ANalysis and TrAnsient Simulation Method (PHANTASM), developed by the University of Wisconsin to predict the energy output of a BIPV installation, requires fewer parameters than the PVFORM/SNL model and most of them are commonly provided by the module manufacturers. The PHANTASM model uses the temperature coefficients and the rating conditions to calculate the series resistance, a model parameter which is not readily available, and the user must input himself the appropriate nominal operating cell temperature of the modules [22].

Like all detailed simulation programs, the above tools are all based upon time-series analysis, typically of hourly values of solar irradiation, wind speed, and ambient temperature, usually covering a one year period. Such versatile programs, however, are not very suitable for web-based applications, which must be characterized by data transfer and execution in real-time, if possible. CPE – the clean power estimator – is a web-based PV economic evaluation program which is based on the realization that average time-of-day and time-of-year PV outputs are sufficient for most economic analyses. Therefore, instead of transferring full yearly sets of data and running full-year simulations, CPE uses average PV output tables pre-calculated by PVFORM. Clearly, specifics like PV operating temperature and temperature response are fixed, i.e. built in the reference tables [23,24].

The building energy programs, which traditionally handle thermal energy and mass flows, eventually supplement their repertoire with PV-related models. The TRNSYS component library, for example, includes routines like Type 70, which models a PV module or array again on the basis of a solar cell's DC electrical model.⁴ For the calculation of cell/module temperature, Type 70 offers four

² Ref. [77] discusses how to determine T_c in the context of NOCT determination for situations where direct temperature measurement is impossible (e.g. in concentrators) or problematic (due to the T-gradient between sensor position and PV cell).

³ A detailed discussion of INOCT can also be found in Ref. [78].

⁴ TRNSYS photovoltaic simulation is combined with Energy-10 load simulation in the "Energy-10 PV" package [79].

modes: T_c is user input, T_c is calculated from a simple energy balance, a more accurate balance gives T_c and, finally, T_c results from an accurate balance on a cell with phase change material backing. In all cases but the first, the program solves an implicit equation for T_c [25]. ESP-r, on the other hand, has based its latest PV component on the WATSUN-PV model [26]. The latter, which has an empirical basis, calculates the short-circuit current and the open-circuit voltage using the actual operating temperature and not just the temperature at reference conditions, as was the case with the previous ESP-r model [27].

Finally, more general platforms like Matlab can be utilized for the development of PV models. For example, a simple PV array simulation model based on reference cell temperature has been recently developed in the Matlab-Simulink GUI environment [28]. On the other hand, a more realistic than the above model featuring temperature dependence in the current components has also been easily implemented in Matlab [29].

5. Conclusions

A key variable for the photovoltaic conversion process is the operating temperature of the cell/module. The numerous correlations for T_c which have appeared in the literature apply to freely mounted PV arrays, to PV/thermal collectors, and to BIPV installations, respectively. They involve basic environmental variables and numerical parameters which are material or system dependent. Thus, one must be careful in applying a particular expression for the operating temperature of a PV module because the available equations have been developed with a specific mounting geometry or building integration level in mind. Therefore, the reader is urged to consult the original sources when seeking a correlation suitable for a particular application.

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